Assignment 7

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MultiJoin: Couse/Student/Enrollment

1. List students by name and ssn who are taking 5 courses.

```
SELECT s.firstName AS First_Name, s.lastName AS "Last_Name", s.ssn AS "SSN"
FROM StudentTable s
INNER JOIN EnrollmentTable e
    ON s.ssn = e.ssn
GROUP BY s.ssn
HAVING count(e.courseId) = 0;
```

First_Name	Last_Name	SSN
Kevin	Green	100000041
Michelle	Adams	100000042
Brian	Nelson	100000043
Carol	Baker	100000044
Edward	Hall	100000045
Ronald	Campbell	100000047
Melissa	Mitchell	100000048
Timothy	Carter	100000049
Deborah	Roberts	100000050
Stephanie	Phillips	100000052
Jeffrey	Evans	100000053
Rebecca	Turner	100000054
Ryan	Diaz	100000055
Laura	Parker	100000056
Sharon	Edwards	100000058
Gary	Collins	100000059
Cynthia	Reyes	100000060

17 rows in set (0.003 sec)

2. List students by name and ssn who are taking 4 courses.

```
HAVING count(e.courseId) = 4;
```

First_Name	Last_Name	SSN
Amanda	Rivera	100000046
Jacob	Cruz	100000057

2 rows in set (0.000 sec)

3. List students by name and ssn who are taking 3 courses.

```
HAVING count(e.courseId) = 3;
```

First_Name	Last_Name	SSN
James	Smith	100000001
Mary	Johnson	100000002
Robert	Williams	100000003
Patricia	Brown	100000004
John	Jones	100000005
Jennifer	Garcia	100000006
Michael	Miller	100000007
Linda	Davis	100000008
David	Rodriguez	100000009
Elizabeth	Martinez	100000010
William	Hernandez	100000011
Barbara	Lopez	100000012
Richard	Gonzalez	100000013
Susan	Wilson	100000014
Joseph	Anderson	100000015
Jessica	Thomas	100000016
Thomas	Taylor	100000017
Sarah	Moore	100000018
Christopher	Jackson	100000019
Karen	Martin	100000020

20 rows in set (0.000 sec)

4. <u>List students by name and ssn who are taking 2 courses.</u>

   HAVING count(e.courseId) = 2;

First_Name	Last_Name	SSN
Nicholas	Stewart	100000061
Kathleen	Morris	100000062
Eric	Morales	100000063
Amy	Murphy	100000064
Stephen	Cook	100000065
Heather	Cole	100000098
Aaron	West	100000099
Diane	Jordan	100000100

39 rows in set (0.000 sec)

5. <u>List students by name and ssn who are taking 1 courses.</u>

First_Name	Last_Name	SSN
Charles	Lee	100000021
Nancy	Perez	100000022
Matthew	Thompson	100000023
Lisa	White	100000024
Anthony	Harris	100000025
Betty	Sanchez	100000026
Mark	Clark	100000027
Margaret	Ramirez	100000028
Donald	Lewis	100000029
Sandra	Robinson	100000030
Steven	Walker	100000031
Ashley	Young	100000032
Paul	Allen	100000033
Dorothy	King	100000034
Andrew	Wright	100000035
Kimberly	Scott	100000036
Joshua	Torres	100000037
Emily	Nguyen	100000038
Kenneth	Hill	100000039
Donna	Flores	100000040

20 rows in set (0.001 sec)

6. List students by name and ssn who are taking 0 courses.
   Empty set (0.000 sec)

7. Give a count of 4 credit courses by subjectID (# of 4-credit courses in each subject).
   SELECT subjectId, COUNT(courseId) AS "# 1-Credit Courses"
   FROM coursetable
   WHERE numOfCredits = 1
   GROUP BY subjectId;

```
+-----------+--------------------+
| subjectId | # 4-Credit Courses |
+-----------+--------------------+
| CS        |                  2 |
| ENGR      |                  2 |
| MATH      |                  2 |
+-----------+--------------------+
3 rows in set (0.001 sec)
```

8. Give a count of 3 credit courses by subjectID
   WHERE numOfCredits = 3

```
+-----------+--------------------+
| subjectId | # 3-Credit Courses |
+-----------+--------------------+
| CS        |                  3 |
| ENGL      |                  5 |
| ENGR      |                  2 |
| MATH      |                  3 |
+-----------+--------------------+
4 rows in set (0.000 sec)
```

9. Give a count of 2 credit courses by subjectID
   WHERE numOfCredits = 2

```
+-----------+--------------------+
| subjectId | # 2-Credit Courses |
+-----------+--------------------+
| ENGR      |                  1 |
+-----------+--------------------+
1 row in set (0.000 sec)
```

10. Are there any 5 credit courses? Are there any 1 credit courses?
    No! They both return “Empty set (0.000 sec)”

11. List all enrolled students by name and their grades in each enrolled course

SELECT s.firstName, s.lastName, e.grade
FROM studenttable AS s
INNER JOIN enrollmenttable AS e ON s.ssn = e.ssn;

```
| Aaron        | West        | B     |
| Diane        | Jordan      | A     |
| Diane        | Jordan      | C     |
+-------------+------------+-------+
251 rows in set (0.000 sec)
```

12. <u>For the course identified by courseID of 10, list the number of students enrolled, course number as well as subjectID.</u>
SELECT c.courseNumber, c.subjectId, COUNT(e.ssn) AS "Total Enrolled"
FROM coursetable AS c
INNER JOIN enrollmenttable AS e ON c.courseId = e.courseId
WHERE c.courseId = 10;

```
+--------------+-----------+----------------+
| courseNumber | subjectId | Total Enrolled |
+--------------+-----------+----------------+
|          450 | CS        |             13 |
+--------------+-----------+----------------+
1 row in set (0.000 sec)
```

13. <u>How many students are in each course (identified uniquely by courseID)?</u>
SELECT courseId, COUNT(ssn) AS "Student Count"
FROM enrollmenttable
GROUP BY courseId;

```
|       17 |            12 |
|       18 |            12 |
|       19 |            11 |
|       20 |            10 |
+----------+---------------+
20 rows in set (0.000 sec)
```

14. <u>For each student enrolled, list the first name, last name, and number of credits that student was taking.</u>
SELECT s.firstName, s.lastName, SUM(c.numOfCredits) AS "Total Credits"
FROM studenttable AS s
INNER JOIN enrollmenttable AS e ON s.ssn = e.ssn
INNER JOIN coursetable AS c ON e.courseId = c.courseId
GROUP BY s.ssn;

```
| Tyler       | Woods      |             8 |
| Heather     | Cole       |             7 |
| Aaron       | West       |             7 |
| Diane       | Jordan     |             6 |
+-------------+------------+---------------+
98 rows in set (0.002 sec)
```